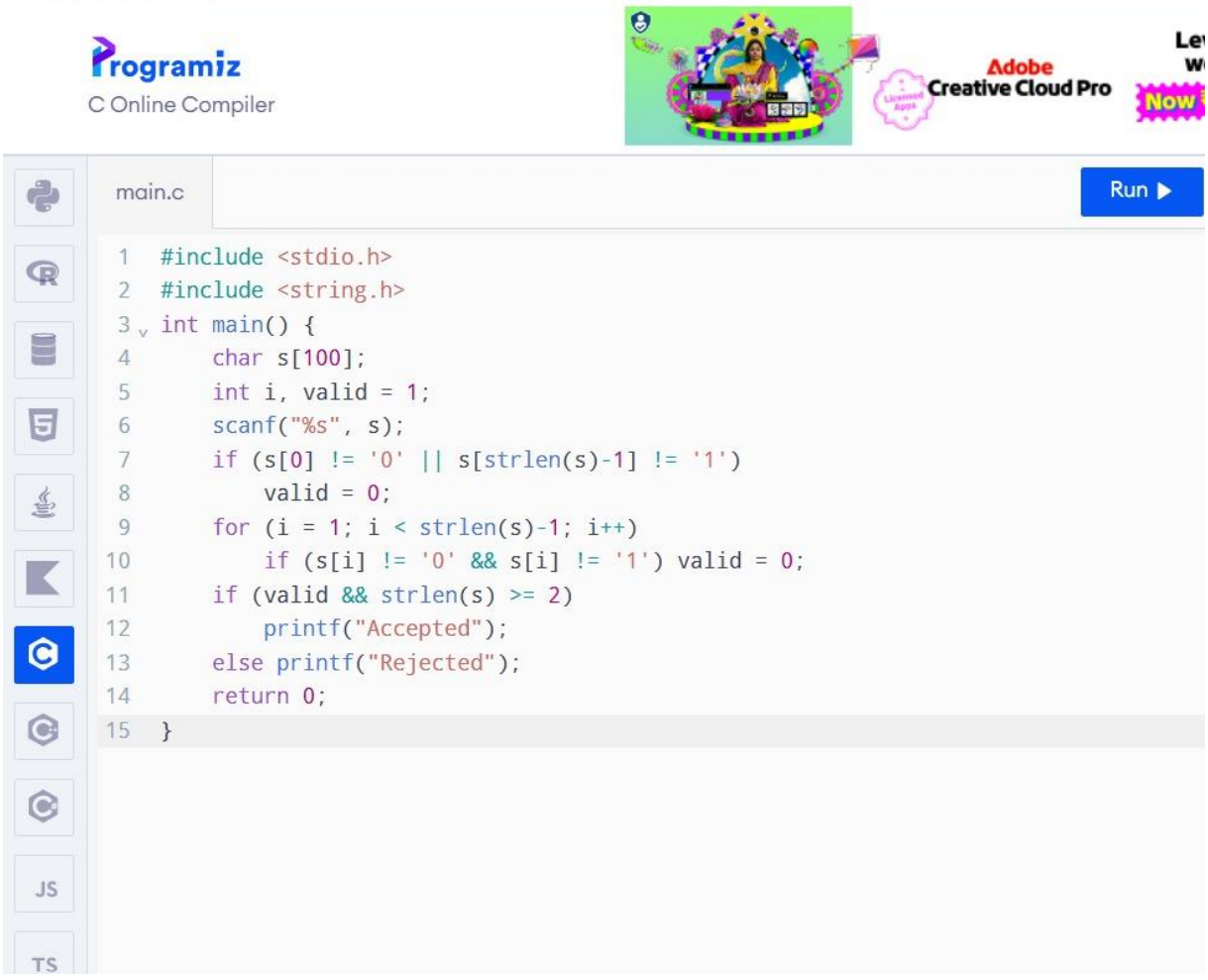
3. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG) S → 0A1 A → 0A | 1A | ε

- Check whether the string starts with 0, ends with 1, and has only binary digits in between.

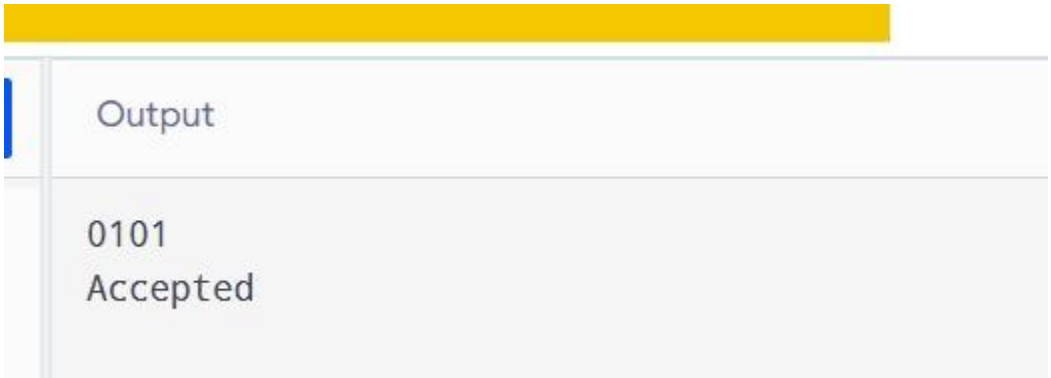
AIM: To write a C program to check whether a string belongs to the given CFG.

CODE:




```c
#include <stdio.h>
#include <string.h>
int main() {
    char s[100];
    int i, valid = 1;
    scanf("%s", s);
    if (s[0] != '0' || s[strlen(s)-1] != '1')
        valid = 0;
    for (i = 1; i < strlen(s)-1; i++)
        if (s[i] != '0' && s[i] != '1') valid = 0;
    if (valid && strlen(s) >= 2)
        printf("Accepted");
    else printf("Rejected");
    return 0;
}
```

Output:



Result: The program accepts strings of the form `0(binary digits)1` and rejects other strings.